Algebra 1 – Module 12 Practice Test

Full Name: ______________________________

Period: _________

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

The daily high temperatures (°F) recorded in Cedar Falls over the last twelve days are given below. Use this data set for problems 1–6.

69, 62, 73, 53, 79, 69, 66, 58, 73, 62, 83, 69

1. Find the mean.

2. Find the median.

3. Find the mode.

4. Find the range.

5. Find the lower quartile (Q1) and the upper quartile (Q3).

6. Find the interquartile range (IQR).

7. Write the five-number summary of this NEW data set: 11, 16, 19, 23, 29, 35, 44

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8. Identify the outlier, and explain how you know: 66, 69, 71, 74, 112

9. For the data 35, 37, 41, 43, 104, find the mean with and without the outlier 104. Describe how the outlier affects the mean.

12. A report lists the number of followers for each student in one class, where one student is a famous influencer. Which measure better represents a TYPICAL value — the mean or the median? Explain your reasoning.

10. Two robotics teams competed in the same three rounds. Their round scores are shown. Show that the two teams have the SAME mean but DIFFERENT ranges.

Team	**Round Scores**
Team Red	68, 74, 80
Team Blue	62, 77, 83

11. Error Analysis — For the data 17, 26, 41, 9, 30, Marcus said the range is 41. Identify his mistake, then find the range.

13. Devin's first three test scores are 74, 85, and 93. What score does he need on his fourth test so that his mean is exactly 87? Show your work.

Cumulative Review — ACT Style. Circle the best answer.

14. What is the vertex of $y = x^2 - 6x + 11$?

Ⓐ (3, 2)
Ⓑ (−3, 2)
Ⓒ (3, 11)
Ⓓ (6, 11)

15. Find the product: $(4x + 1)(x + 3)$

Ⓔ $4x^2 + 13x + 3$
Ⓕ $4x^2 + 12x + 3$
Ⓖ $4x^2 + 13x + 4$
Ⓗ $5x + 4$

16. Which ordered pair solves the system $y = x + 5$ and $x + y = 11$?

Ⓐ (3, 8)
Ⓑ (8, 3)
Ⓒ (3, 3)
Ⓓ (4, 9)